

Physics Assignment

Review paper for Midterm Exam

1. Evaluate i) $\frac{25.2 \times 1374}{33.3}$ 3 s.f.

$$= \frac{34624.8}{33.3} = 1039.783$$

$$= 1039.7$$

$$= 1040 = 0.104 \times 10^4$$

ii) $24.36 + 0.0623 + 256.2$ 1 dp

$$= 24.4223 + 256.2$$

$$= 280.62$$

$$= 280.6 \quad 1 \text{ dp}$$

2. Newton's law of universal gravitation is represented by $F = \frac{GMm}{r^2}$, where F is the force, M & m are masses of the two objects and r is the centre to centre distance between the two objects. Find the dimension of G and hence state its SI unit.

$$F = \frac{GMm}{r^2}$$

$F \rightarrow$ Force $G \rightarrow$ Gravitation constant
 $M \& m \rightarrow$ masses $r \rightarrow$ distance

$$G = \frac{Fr^2}{Mm}$$

$$\text{Force } [M][L][T]^{-2}$$

$$\text{Distance squared } [L]^2$$

$$\text{Product of masses } [M]^2$$

→ Dimension of G

$$\Rightarrow \frac{[M][L][T]^{-2} [L]^2}{[M]^2}$$

$$\Rightarrow [M]^{-1} [L]^3 [T]^{-2}$$

SI unit of G

$$M \rightarrow \text{kg} \quad L \rightarrow \text{m} \\ T \rightarrow \text{s} \\ \text{kg}^{-1} \text{m}^3 \text{s}^{-2}$$

- c3. When a spherical object moves through a liquid, the liquid opposes the motion with a force F . The magnitude of F depends on the coefficient of viscosity, η of the liquid (whose dimensions are $ML^{-1}T^{-1}$), the speed, v of the object & the radius, r of the object. Assuming that F is proportional to different powers of these quantities, guess a formula for F using the method of dimensions.

3. F is assumed to be proportional ~~to~~
to powers of η, v, r ~~then~~

$$F \propto \eta^a v^b r^c$$

$$F = k \cdot \eta^a v^b r^c$$

Let k be the dimensionless constant &
 a, b, c are exponents that we have to find

Quantity	Symbol	Dimension
Force	F	$[M][L][T]^{-2}$
Coefficient of viscosity	η	$[M][L]^{-1}[T]^{-1} \rightarrow \text{Given}$
Speed	v	$[L][T]^{-1}$
Radius	r	$[L]$

$$[M][L][T]^{-2} = ([M][L]^{-1}[T]^{-1})^a ([L][T]^{-1})^b ([L])^c$$

$$[M][L][T]^{-2} = [M]^a [L]^{-a+b+c} [T]^{-a-b}$$

$$\begin{array}{ccc} \text{from } [M] & \text{from } [L] & \text{from } [T] \end{array}$$

$$a = 1$$

$$-a + b + c = 1$$

$$-a - b = -2$$

$$-1 + b + c = 1$$

$$a + b = 2$$

$$b + c = 2$$

$$b = 2 - 1 = 1$$

$$c = 2 - 1 = 1$$

$$a = b = c = 1$$

$$\therefore F \propto r^n \cdot v^m \cdot r^p$$

$$F = k r^n v^m r^p$$

$\therefore$ The guessed formula for Force is

$$F = K r^n v^m r^p$$

4. The position of a particle (x) moving along the x -axis as function of time (t) is given by $x = at^3 + bt^2 + ct + d$. The numerical values of a, b, c and d are 1, 4, -2 & 5 respectively & SI units are used for x & t . Find:

- the SI unit of a, b, c & d
- the velocity of the particle at $t = 4s$
- the acceleration of the particle at $t = 4s$
- the average velocity between $t = 0$ & $t = 4s$

$$x = at^3 + bt^2 + ct + d$$

$$\text{SI unit of } x = m \quad \text{SI unit of } t = s$$

- $at^3 = a$ has units $\frac{m}{s^3}$
 $bt^2 \Rightarrow b$ has units $\frac{m}{s^2}$
 $ct \Rightarrow c$ has units $\frac{m}{s}$
 $d \Rightarrow d$ has units m

ii) Velocity at $t = 4$ s

$$v(t) = \frac{dx}{dt} = 3at^2 + 2bt + c$$

$$\begin{aligned} v(4) &= 3(1)(4)^2 + 2(4)(4) + (-2) \\ &= 3 \times 16 + 2(16) - 2 \\ &= 48 + 32 - 2 = 48 + 30 \\ &= 78 \text{ m/s} \end{aligned}$$

iii) Acceleration at $t = 4$ s

$$a_{\text{inst}}(t) = \frac{dv}{dt} = 6at + 2$$

$$\begin{aligned} a_{\text{inst}}(4) &= 6(1)(4) + 2(4) = 24 + 8 \\ &= 32 \text{ m/s}^2 \end{aligned}$$

iv) Average velocity between $t = 0$ & $t = 4$ s

~~Average~~ $x(4) = 1 \cdot 4^3 + 4 \cdot 4^2 - 2 \cdot 4 + 5$
 $= 64 + 64 - 8 + 5 = 125 \text{ m}$

$$x(0) = d = 5 \text{ m}$$

$$v_{\text{avg}} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{125 - 5}{4 - 0} = \frac{120}{4} = 30 \text{ m/s}$$

- 5) A particle moves in the X-Y plane with a constant acceleration of 1.5 m/s^2 in the direction making an angle of 37° with the x-axis. At $t=0$ the particle is at the origin & its velocity is 8.0 m/s along the x-axis. Find:
- the velocity of the particle at $t=4$
 - the position of the particle at $t=4.0 \text{ s}$

Given: constant acceleration $a = 1.5 \text{ m/s}^2$

Angle of acceleration with x-axis: $\theta = 37^\circ$

initial velocity: $u = 8.0 \text{ m/s}$

$$\Rightarrow u_x = 8.0 \text{ m/s} \quad u_y = 0$$

Time $t = 4 \text{ s}$

$$a_x = a \cos \theta = 1.5 \cos 37^\circ = 1.191 = 1.2 \text{ m/s}^2$$

$$a_y = a \sin \theta = 1.5 \sin 37^\circ = 0.902 = 0.9 \text{ m/s}^2$$

$$V_x = V_{x0} + a_x t = 8.0 + (1.2)(4) = 8 + 4.8 = 12.8 \text{ m/s}$$

$$V_y = V_{y0} + a_y t = 0 + (0.9)(4) = 3.6 \text{ m/s}$$

i) magnitude of velocity $V = \sqrt{V_x^2 + V_y^2} = \sqrt{(12.8)^2 + (3.6)^2}$

$$= \sqrt{163.84 + 12.96} = \sqrt{176.8}$$

$$\approx 13.3 \text{ m/s}$$

Direction $\Rightarrow \tan \phi = \frac{V_y}{V_x} = \frac{3.6}{12.8} = 0.281$

$$\phi = \tan^{-1}(0.281) \approx 15.7^\circ$$

velocity of the particle at $t = 4\text{ s}$

$\Rightarrow 13.3\text{ m/s}$ at 15.7° above x -axis

$$\begin{aligned} \text{ii) } x &= v_x t + \frac{1}{2} a_x t^2 = 8(4) + \frac{1}{2} (1.2) (4^2) \\ &= 32 + \frac{19.2}{2} \\ &= 32 + 9.6 = 41.6\text{ m} \end{aligned}$$

$$y = v_y t + \frac{1}{2} a_y t^2 = 0 + \frac{1}{2} (0.9) (16) = \frac{14.4}{2} = 7.2$$

Position at $t = 4\text{ s}$

$$(x, y) = (41.6\text{ m}, 7.2\text{ m})$$

magnitude of displacement:

$$\begin{aligned} r &= \sqrt{x^2 + y^2} = \sqrt{(41.6)^2 + (7.2)^2} = \sqrt{1730.56 + 51.84} \\ &= \sqrt{1782.4} \approx 42.2\text{ m} \end{aligned}$$

$$\tan \theta = \frac{y}{x} = \frac{7.2}{41.6} = 0.173$$

$$\theta = \tan^{-1}(0.173) \approx 9.9^\circ$$

position of the particle at $t = 4\text{ s}$
 $\Rightarrow 42.2\text{ m}$ at 9.9°

- 6) A stone is thrown vertically upward from a point on a bridge located 40 m above the water. Knowing that it strikes the water 4 s after release, determine
- the speed with which the stone was thrown upward
 - the speed with which the stone strikes the water

Given : Height of bridge above water $h = 40 \text{ m}$

Time of flight $t = 4 \text{ s}$

Acceleration due to gravity : $g = 9.8 \text{ m/s}^2$

The stone is thrown vertically upward from the bridge

i) $s = ut + \frac{1}{2}gt^2$

$$-40 = u(4) + \frac{1}{2}(-9.8)(4)^2$$

$$-40 = u(4) - 4.9(16)$$

$$4u = 78.4 - 40$$

$$u = \frac{38.4}{4} = 9.6$$

-ve because
it ends up 40 m
below starting point

$u = 9.6 \text{ m/s}$ upward

speed with which the stone was thrown upward $\Rightarrow u = 9.6 \text{ m/s}$ upward

ii) Speed with which the stone strikes the water

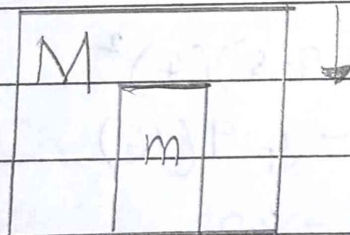
$$\begin{aligned} v &= u - gt \\ &= 9.6 - 9.8(4) \\ &= 9.6 - 39.2 = -29.6 \end{aligned}$$

-ve sign means downward

magnitude of speed $|v|$

$\Rightarrow 29.6 \text{ m/s downward}$

7. A box of mass M contains an object of mass m as shown in the below. What should be the acceleration of the box during its descent if the object exerts a force of $mg/4$ on the floor of the box?



mass of object = ~~m~~ mass of box = M

downward acceleration (since box descend)

object exerts a force of $mg/4$ on the floor of the box

$$\Rightarrow F = \frac{mg}{4}$$

Forces on the object

Weight acting downward = mg

Normal acts upward = N

given $N = \frac{mg}{4}$

Net force on object = ma

$$mg - N = ma$$

$$\left(N = \frac{mg}{4} \right) \quad mg - \frac{mg}{4} = ma$$

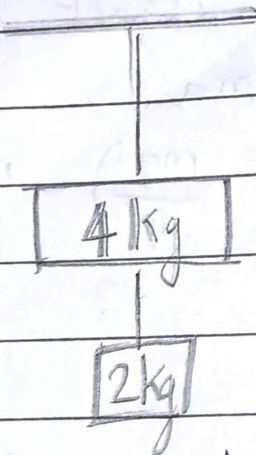
$$\frac{3mg}{4} = ma$$

$$a = \frac{3g}{4} = \frac{3 \times 9.8}{4}$$

$$= \frac{29.4}{4} = 7.35 \text{ m/s}^2$$

Acceleration of the box during its descent
 $\Rightarrow 7.35 \text{ m/s}^2$

8. A 4 kg block is suspended from the roof of an elevator. A 2 kg block is suspended from the 4 kg block (see fig below). When the elevator accelerates upwards 2.2 m/s^2 , find the magnitudes of the tensions in the strings.



$$m_1 = 4 \text{ kg} \quad m_2 = 2 \text{ kg}$$

$$a = 2.2 \text{ m/s}^2 \quad g = 9.8 \text{ m/s}^2$$

For the lower block mass m_2 upward tension $T_2 - m_2 g = m_2 a =$ (elevator accelerates upward)

$$T_2 = m_2 (g + a)$$

$$= 2 (2.2 + 9.8)$$

$$= 2 (12) = 24 \text{ N}$$

For the top block mass m_1 , top tension T_1 supports both block masses so

$$T - (m_1 + m_2)g = (m_1 + m_2)a$$

$$T_1 = (m_1 + m_2)a + (m_1 + m_2)g$$

$$= (m_1 + m_2)(a + g)$$

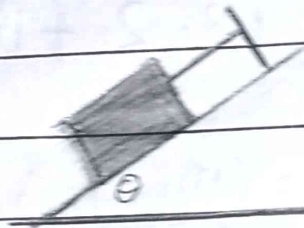
$$= (4 + 2)(2.2 + 9.8)$$

$$= 6 \times 12.0 = 72 \text{ N}$$

Magnitude of the two strings

$\Rightarrow$ Hence $T_2 = 24 \text{ N}$, $T_1 = 72 \text{ N}$

9. A block of mass 42 kg rests on a slope with a smooth surface. It is tied by a string to a point. The string can withstand a maximum tension of 300 N . Determine the angle at which the string is about to snap.



mass of block $= m = 42\text{ kg}$

Maximum tension in the string, $T_{\text{max}} = 300\text{ N}$

Surface is smooth (no friction)

Acceleration due to gravity $g = 9.8\text{ m/s}^2$

Forces on the block

vertically downward weight mg

Normal - perpendicular to the slope

Tension T acting up the slope

$$W = mg \sin \theta$$

$$T = mg \sin \theta$$

$$T_{\max} = mg \sin \theta_{\max}$$

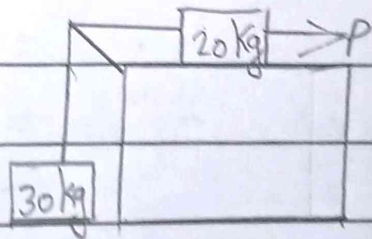
$$\begin{aligned} \sin \theta_{\max} &= \frac{T_{\max}}{mg} \\ &= \frac{300}{42 \times 9.8} = \frac{300}{411.6} = 0.7288 \end{aligned}$$

$$\sin \theta_{\max} = 0.7288$$

$$\theta_{\max} = \sin^{-1}(0.7288) = 46.78^\circ \approx 46.8^\circ$$

$$\theta_{\max} = 47^\circ$$

10. A force P is acting on block B as shown. If $P = 200\text{ N}$, find the tension of the string and the acceleration of the system.



$$m_A = 30\text{ kg}$$

$$m_B = 20\text{ kg}$$

$$P = 200\text{ N}$$

$$g = 9.8\text{ m/s}^2$$

$$P - T = m_B a \Rightarrow T = P - m_B a$$

$$T - m_A g = m_A a$$

$$T = m_A a + m_A g = m_A (a + g)$$

$$P - T = m_B a$$

$$+ \cancel{T} - m_A g = m_A a$$

$$P - m_A g = m_B a + m_A a$$

$$200 - 30(9.8) = a(30 + 20)$$

$$\frac{-94}{50} = a$$

$$a = -1.88 \text{ m/s}^2$$

-ve sign means Block A moves down
block B moves left

magnitude of acceleration

$$\Rightarrow a \Rightarrow \cancel{1.88} 1.88 \text{ m/s}^2$$

$$T - m_A g = m_A a$$

$$T = m_A a + m_A g$$

$$= m_A (a + g)$$

$$= 30(-1.88 + 9.8)$$

$$= \cancel{30(11.68)} 30(7.92)$$

$$= \cancel{350.4} 237.6 \text{ N}$$

$$T \approx 238 \text{ N}$$

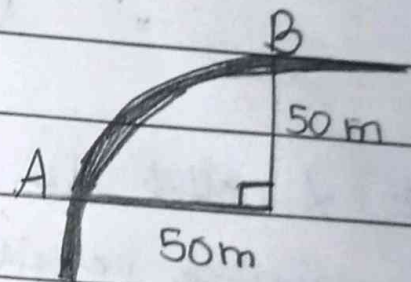
11. A car of mass 1000 kg is travelling along a road at a constant speed of 72 kmh^{-1} . It enters a bend at A and exits at B. The bend has a radius of 50 m . (a) What is the magnitude of the centripetal force that is experienced by the car? b) If your answer in (a) is the largest force that the car can withstand, what is the maximum speed of the car if the radius is 32 m instead?

$$m = 1000 \text{ kg}$$

$$\text{speed} = 72 \text{ km/h}$$

$$v = 72 \times \frac{1000}{3600} = 20 \text{ m/s}$$

$$\text{radius } r = 50 \text{ m}$$



(a) Magnitude of centripetal force

$$F_c = \frac{mv^2}{r}$$

$$= \frac{1000 \times (20)^2}{50} = \frac{4 \times 10^4}{5}$$

$$= 0.8 \times 10^4 = 8000 \text{ N}$$

$$F_c = 8000 \text{ N}$$

b) Maximum speed for $r_2 = 32\text{m}$ when $F_c = 8000$

$$F_c = \frac{mv^2}{r}$$

$$8000 = \frac{1000 (V)^2}{32}$$

$$\frac{8000 \times 32}{1000} = V^2$$

$$V = \sqrt{256} = 16 \text{ m/s}$$

$$V_{\text{max}} = 16 \text{ m/s} \text{ or } 57.6 \text{ km/h}$$

12. A clock has its minute hand 4.0cm long. Find the average velocity of the tip of the minute hand

(a) Between $6:00\text{am}$ to $6:30\text{am}$ &

(b) Between $6:00\text{am}$ to $6:30\text{pm}$.

length of minute hand $r = 4\text{cm}$

average velocity = ?

(a) At 6 the minute hand points at 12 (straight up)
At 6:30 the minute hand points at 6 (straight down)

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So, the tip has moved 180° (π radians) on a circle of radius 4 cm

$$\text{Displacement} = 2r = 2 \times 4 = 8 \text{ cm}$$

$$\text{Time interval} = 30 \text{ min} = 1800 \text{ s}$$

$$V_{\text{avg}} = \frac{8}{1800} = 4.4 \times 10^{-3} \text{ cm/s downward}$$

b) ~~between~~ between 6 am & 6:30 pm

$$12 \text{ hr } 30 \text{ min} = 12.5 \text{ hr}$$

after 12 hrs the hand is at 12 o'clock position
after 30 min it is at 6 o'clock position as in part (a)

hence initial position 6 am & final position 6:30 pm are same as before

$$\text{displacement} = 8 \text{ cm} \quad \text{time} = 12.5 \text{ h} = 12.5 \times 3600 = 45000 \text{ s}$$

$$V_{\text{avg}} = \frac{8}{45000} = 1.77 \times 10^{-4} \text{ cm/s downward}$$

14) The sum of two vectors, p & q is perpendicular to p . Find the angle between p and q .

$$p+q \perp p \therefore p \cdot (p+q) = 0$$

$$p \cdot p + p \cdot q = 0$$

$$p^2 + pq \cos \theta = 0$$

$$pq \cos \theta = -p^2$$

$$\cos \theta = \frac{-p^2}{pq}$$

$$\cos \theta = -\frac{p}{q}$$

↓ since $\cos \theta$ -ve it means that the two vectors are more than 90° apart

* A particle travelled between two points. It travels half the distance with a velocity v_0 while the remaining part of the distance was covered with velocity v_1 for half the time and with velocity v_2 for the other half of the time. Find the average velocity during the journey. (Assume the particle travels in a straight line).

13.

let total distance travelled by the particle be D
 first half of distance $\frac{D}{2}$ with velocity v_0
 second half of the distance $\frac{D}{2}$ such that
 half of the time with velocity v_1 & other
 half of the time with velocity v_2

Time for the first half of distance

$$t_1 = \frac{\text{distance}}{\text{velocity}} = \frac{D/2}{v_0} = \frac{D}{2v_0}$$

Distance covered in the second half
 total time $- t_1$

$$t_1 = t_2 = \frac{t_2}{2}$$

$$d_1 = v_1 \frac{t_2}{2}$$

$$d_2 = v_2 \frac{t_2}{2}$$

$$d_1 + d_2 = \frac{t_2}{2} (v_1 + v_2)$$

$$\frac{D}{2} = \frac{t_2}{2} (v_1 + v_2)$$

$$t_2 = \frac{D}{v_1 + v_2}$$

$$t_{\text{total}} = t_1 + t_2 = \frac{D}{2v_0} + \frac{D}{v_1 + v_2}$$

$$V_{avg} = \frac{\text{total Distance}}{\text{total time}} = \frac{D}{t_{total}}$$

$$= \frac{D}{\frac{D}{2V_0} + \frac{D}{V_1 + V_2}} = \frac{D(2V_0(V_1 + V_2))}{D(V_1 + V_2) + D(2V_0)}$$

$$V_{avg} = \frac{2V_0(V_1 + V_2)}{V_1 + V_2 + 2V_0}$$

15. An aircraft is flying horizontally with speed v at altitude h . If it drops a packet to be collected by a man standing on the ground, at what distance from the man should the packet be dropped? (The man stands in the vertical plane of the aircraft's motion)

Given: Aircraft flying horizontally with constant speed v
altitude (height) of flight = h

Packet is dropped & falls freely under gravity

The man is standing vertically below the aircraft's path

When the packet is dropped its initial vertical velocity = 0
(Since its moving horizontally).

Vertical motion under gravity $g \Rightarrow h = \frac{1}{2}gt^2$
 $\therefore t = \sqrt{\frac{2h}{g}}$

During this time, the packet also moves horizontally with the same horizontal velocity as the aircraft (because no horizontal acceleration). $x = v \times t$

$$x = (v) \times \left(\sqrt{\frac{2h}{g}} \right)$$

$$x = v \sqrt{\frac{2h}{g}}$$

16. A 50-kg box, starting from rest, is pulled for 2m across a floor with a constant horizontal force of 200N. The coefficient of kinetic friction is 0.03.
- (a) What is the net force acting on the object?
- (b) Find the final velocity of the box.

Given: mass of box $m = 50 \text{ kg}$

Applied force $F = 200 \text{ N}$

Coefficient of kinetic friction $\mu_k = 0.03$

Distance $s = 2 \text{ m}$ Initial velocity $u = 0$

Acceleration due to gravity $g = 9.8 \text{ m/s}^2$

$$(a) \quad f_k = \mu_k N = \mu_k mg = 0.03 \times 50 \times 9.8 \\ = 1.5 \times 9.8 = \cancel{14.7} \quad 14.7 \text{ N}$$

$$F_{\text{net}} = F - f_k = 200 - 14.7 = 185.3 \text{ N}$$

$$F_{\text{net}} \Rightarrow \text{Net force acting on the object} = 185.3 \text{ N}$$

$$(b) \quad a = \frac{F_{\text{net}}}{m} = \frac{185.3}{50} = 3.706 \text{ m/s}^2$$
$$v^2 = u^2 + 2as = 0 + 2(3.706)2$$
$$v = \sqrt{14.824} = 3.85 \text{ m/s}$$

Final Velocity of the box = 3.85 m/s

17. A 35 kg child swings to and fro on a swing supported by two chains each 3 m long. The tension in each chain at the lowest point of the motion is 280 N. At the lowest point, find the
- force exerted by the seat on the child (Ignore the mass of the seat).
 - Child's speed assuming that it does not hold the chain at the lowest point.

$$T = 280 \text{ N} \quad L = 3 \text{ m} \quad m = 35 \text{ kg} \quad g = 9.8 \text{ m/s}^2$$

- (a) upward force equals to the each Tension so

$$\therefore F_{\text{seat}} = 2T$$

$$= 2 \times 280 = 560 \text{ N upward}$$

(b) $F_{\text{net}} = 2T - mg = 560 - 35(9.8)$

$$= 560 - 343 = \cancel{15} \text{ N } 217 \text{ N}$$

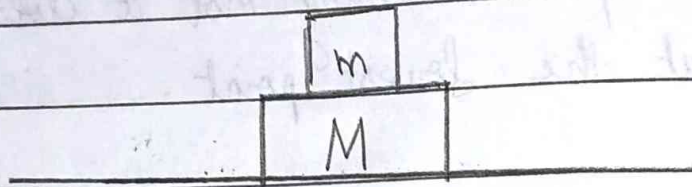
$$a_c = v^2 / L \quad a_c = \frac{217}{m} = \frac{217}{35} = 6.2 \text{ m/s}^2$$

$$6.2 = \frac{v^2}{3}$$

$$v^2 = 18.6$$

$$v = \sqrt{18.6} = 4.31 \text{ m/s}$$

18 In the figure below a ~~big~~ block of mass M lies on a surface that is smooth. A smaller block of mass m lies on the bigger block. The coefficient of static friction between the two blocks is μ . What is the maximum horizontal force that can be applied to the bigger block so that the blocks move together?



mass of big block $\rightarrow M$ mass of small block $\rightarrow m$
static friction between blocks $\rightarrow \mu$

When F (horizontal) is applied to big block both blocks accelerate together with acceleration a .
frictional force between them keeps the small block from slipping

$$F = ma$$

$$f_{\text{max}} = \mu mg$$

$$ma \leq \mu mg$$

$$a_{\text{max}} = \mu g$$

Applying Newton's second law

$$F = (M+m) a_{\max} \\ = (M+m) \mu g$$

$$\therefore F_{\max} = (M+m) \mu g$$

19. (a) In the below equation, the SI units of x & x_0 are in metres, t & t_0 are in seconds, v_0 is m/s & a is m/s^2 . Using the dimensional analysis show whether the equation is homogenous or not.

$$x = x_0 + v_0 (t + t_0) + \frac{1}{2} a (t - t_0)^2$$

- b) A particle is under a constant force

$\mathbf{f} = (-2\mathbf{i} + 4\mathbf{j} + \mathbf{k})\text{N}$ as it moves from a point $(5, 1, 2)\text{m}$ to another point $(-2, 3, 4)\text{m}$. Find the work done on the particle

- c) What is the angle between the displacement and force vectors in (b)?

$$(a) \quad x - x_0 = v_0(t - t_0) + \frac{1}{2} a (t - t_0)^2$$

Dimensions

$$\text{LHS } [x - x_0] = \text{length} = L$$

$$\text{RHS } v_0(t - t_0) \rightarrow [v_0] = LT^{-1}$$

$$\rightarrow [t - t_0] = T$$

$$[v_0(t - t_0)] \rightarrow LT^{-1} \times T = L$$

$$\frac{1}{2} a (t - t_0)^2$$

$$\rightarrow [a] = LT^{-2}$$

$$\rightarrow [(t - t_0)^2] = T^2$$

$$\left[\frac{1}{2} a (t - t_0)^2 \right] = LT^{-2} \times T^2 = L$$

$$\left[v_0(t - t_0) + \frac{1}{2} a (t - t_0)^2 \right] = L + L = L$$

LHS = RHS dimensionally

$\therefore$ the equation is dimensionally homogeneous.

$$b) \vec{F} = (-2\hat{i} + 4\hat{j} + 1\hat{k}) \text{ N}$$

initial point : A (5, 1, 2)

final point : B (-2, 3, 4)

$$\text{Displacement : } \vec{r} = \vec{r}_B - \vec{r}_A = (-2-5)\hat{i} + (3-1)\hat{j} + (4-2)\hat{k} \\ = -7\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\text{Work done} = \vec{F} \cdot \vec{r} = (-2\hat{i} + 4\hat{j} + 1\hat{k}) \cdot (-7\hat{i} + 2\hat{j} + 2\hat{k}) \\ = 14 + 8 + 2 = 24 \text{ J}$$

$$c) \vec{F} \cdot \vec{r} = |\vec{F}| |\vec{r}| \cos \theta$$

$$\cos \theta = \frac{\vec{F} \cdot \vec{r}}{|\vec{F}| |\vec{r}|}$$

$$= \frac{24}{\sqrt{21} \times \sqrt{57}}$$

$$= \frac{24}{\sqrt{1197}}$$

$$= 24 = 0.69$$

$$34.59$$

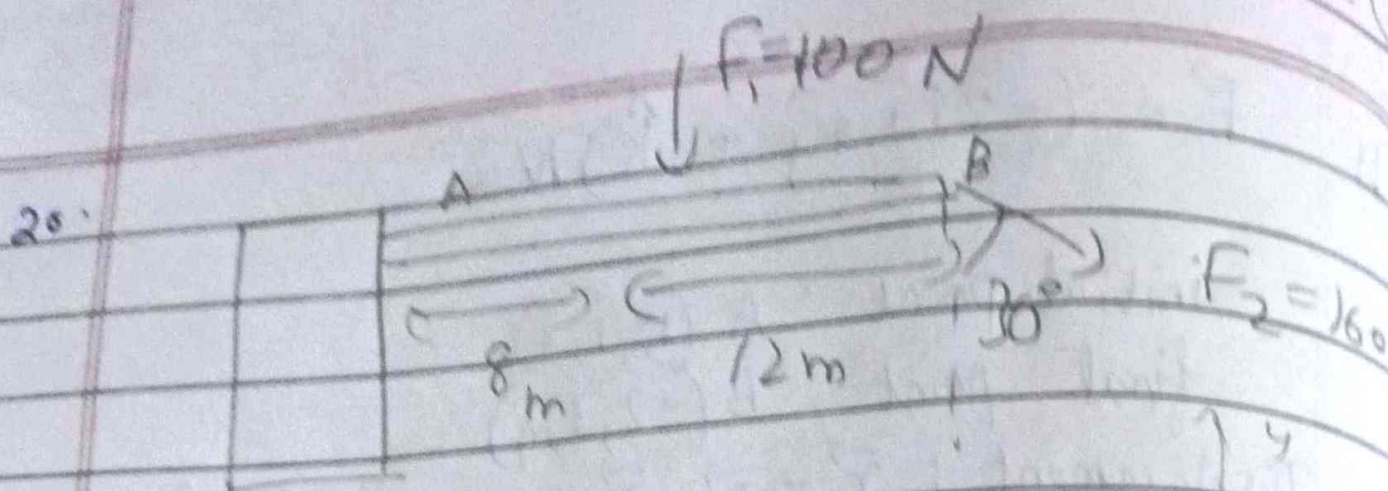
$$\theta = \cos^{-1}(0.69)$$

$$= 46.369^\circ$$

$$\approx 46^\circ$$

$$|\vec{F}| = \sqrt{2^2 + 4^2 + 1^2} \\ = \sqrt{4 + 16 + 1} \\ = \sqrt{21}$$

$$|\vec{r}| = \sqrt{7^2 + 2^2 + 2^2} \\ = \sqrt{49 + 4 + 4} \\ = \sqrt{57}$$



$$F_1 = 100\text{ N} \quad F_2 = 160\text{ N}$$

$$A \text{ to } F_1 = 8\text{ m}$$

$$A \text{ to } B = 8 + 12 = 20\text{ m}$$

(a) $\vec{F}_1$ in terms of unit vector
 $\vec{F}_1$ acts vertically downwards
 $\vec{F}_1 = -100\hat{j}\text{ N}$

$\vec{F}_2$ in terms of unit vector
 $\vec{F}_2 \rightarrow$ magnitude $\rightarrow 160\text{ N}$
 direction - 30° below downward
 (to the right)

$$\vec{F}_2 = (160 \cos(30^\circ))\hat{i} - (160 \sin(30^\circ))\hat{j}$$

$$= 138.6\hat{i} - 80\hat{j}$$

b) Torque about point A

$$\tau = \vec{r} \times \vec{F}$$

for force $\vec{F}_1$

$$\vec{r}_1 = 8\hat{i} \text{ m}$$

$$\vec{F}_1 = -100\hat{j} \text{ N}$$

$$\tau_1 = \vec{r}_1 \times \vec{F}_1$$

$$= 8\hat{i} \text{ m} \times (-100)\hat{j} \text{ N}$$

$$= -800\hat{k} \text{ Nm}$$

for force $\vec{F}_2$

$$\vec{r}_2 = 20\hat{i} \text{ m}$$

$$\vec{F}_2 = 138.6\hat{i} - 80\hat{j} \text{ N}$$

$$\tau = \vec{r}_2 \times \vec{F}_2$$

$$= 20\hat{i} \text{ m} \times 138.6\hat{i} - 80\hat{j} \text{ N}$$

$$= 0 + 1600\hat{k}$$

$$= 1600\hat{k}$$